Automated Re-Training Pipeline Triggered by Data Drift

Assignment 8 | Demand Forecasting | MLOps PBA2

Student Details

Name:	Tanmay Shinde
Subject:	MLOps PBA2
Enrollment No.:	ADT23SOCB1617
Class:	LY-SMAD
GitHub:	github.com/tanmays0/automated-drift-retraining-pipeline

Problem Statement

Domain: Demand Forecasting
Tech Stack: Prefect or Airflow, Evidently AI, Scikit-Learn, MLflow

Build an automated pipeline that continuously monitors incoming data streams for data and concept drift using Evidently AI. When drift exceeds a defined threshold (e.g., Wasserstein distance > 0.1), automatically trigger a re-training DAG.

Key Tasks:
1. Generate synthetic drifted data.
2. Register newly trained models into an MLflow Registry.
3. Set up alert notifications (Email).

Tech Stack (Implemented)

Prefect | Evidently AI | Scikit-Learn | MLflow | Email alerts (local files)

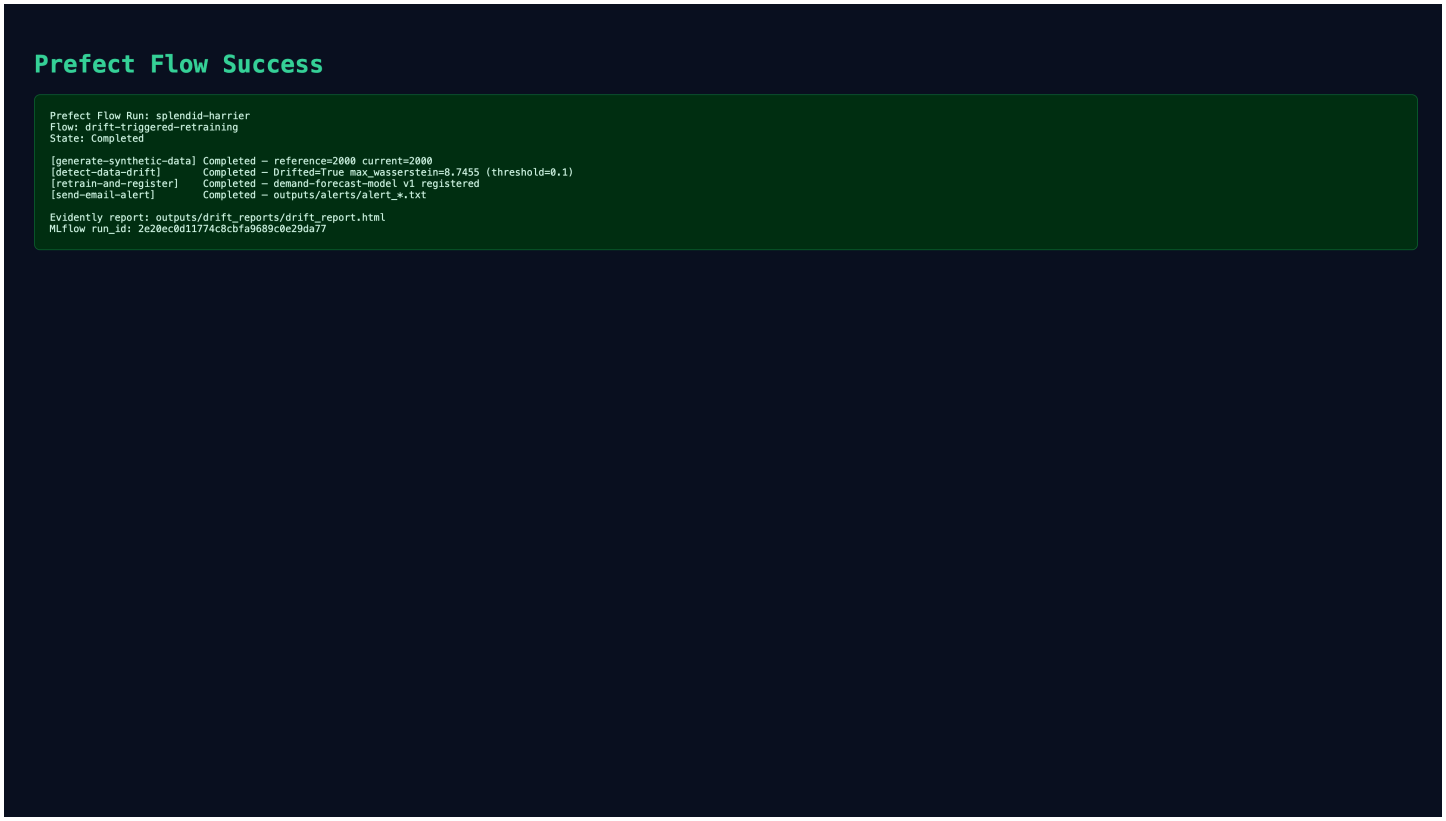
Key Results

- Drift detected: True
- Max Wasserstein distance: 8.7455 (threshold 0.1)
- Model registered: demand-forecast-model v1
- Metrics: MAE ~ 3.52, R2 ~ 0.95
- Alert: email notification written to outputs/alerts/

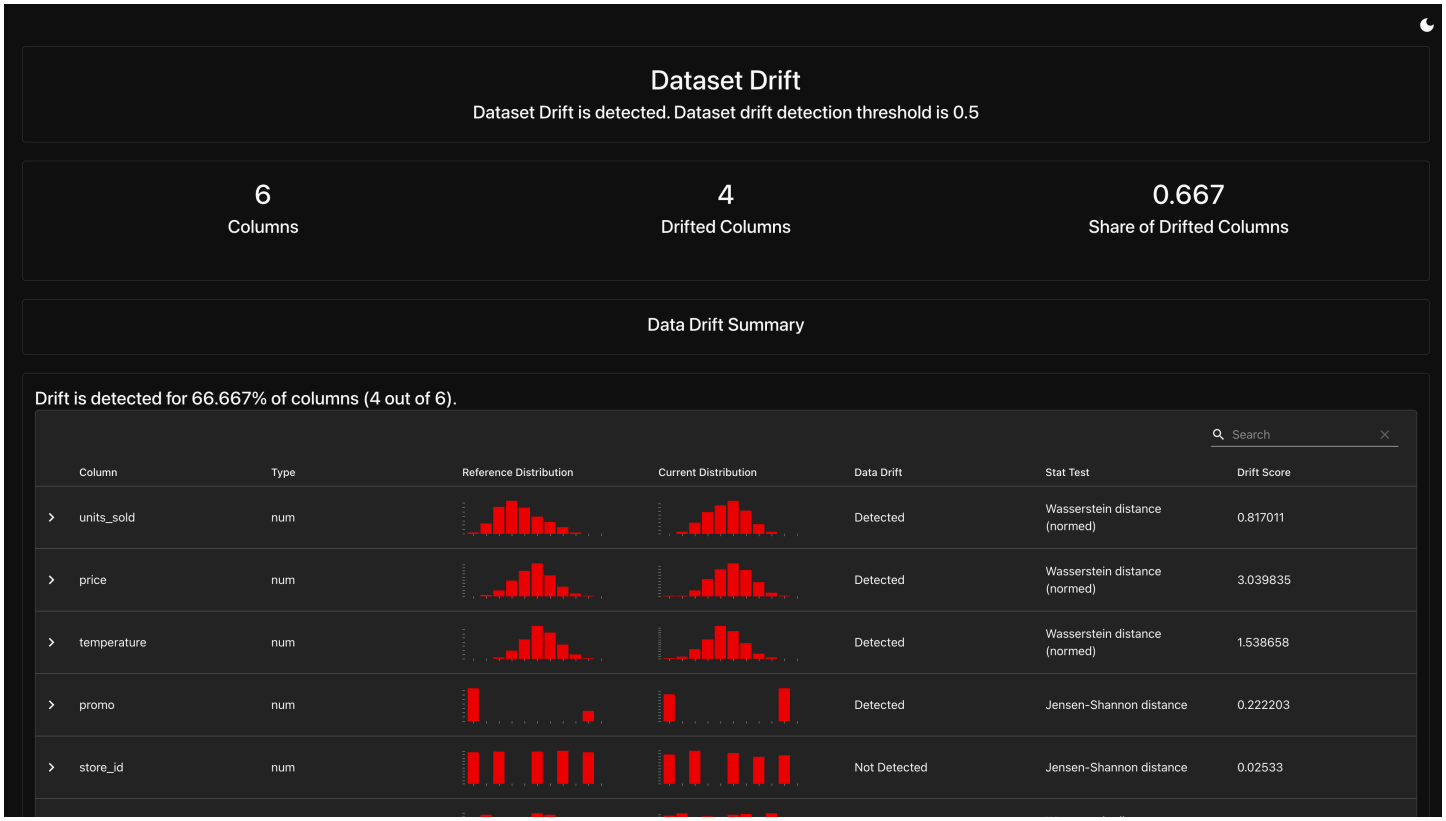
Where to check live output

GitHub: https://github.com/tanmays0/automated-drift-retraining-pipeline
MLflow UI: http://127.0.0.1:5000/#/models
Evidently report: http://127.0.0.1:8765/drift_report.html (or outputs/drift_reports/drift_report.html)
Email alert: outputs/alerts/alert_*.txt
Re-run: python scripts/run_demo.py

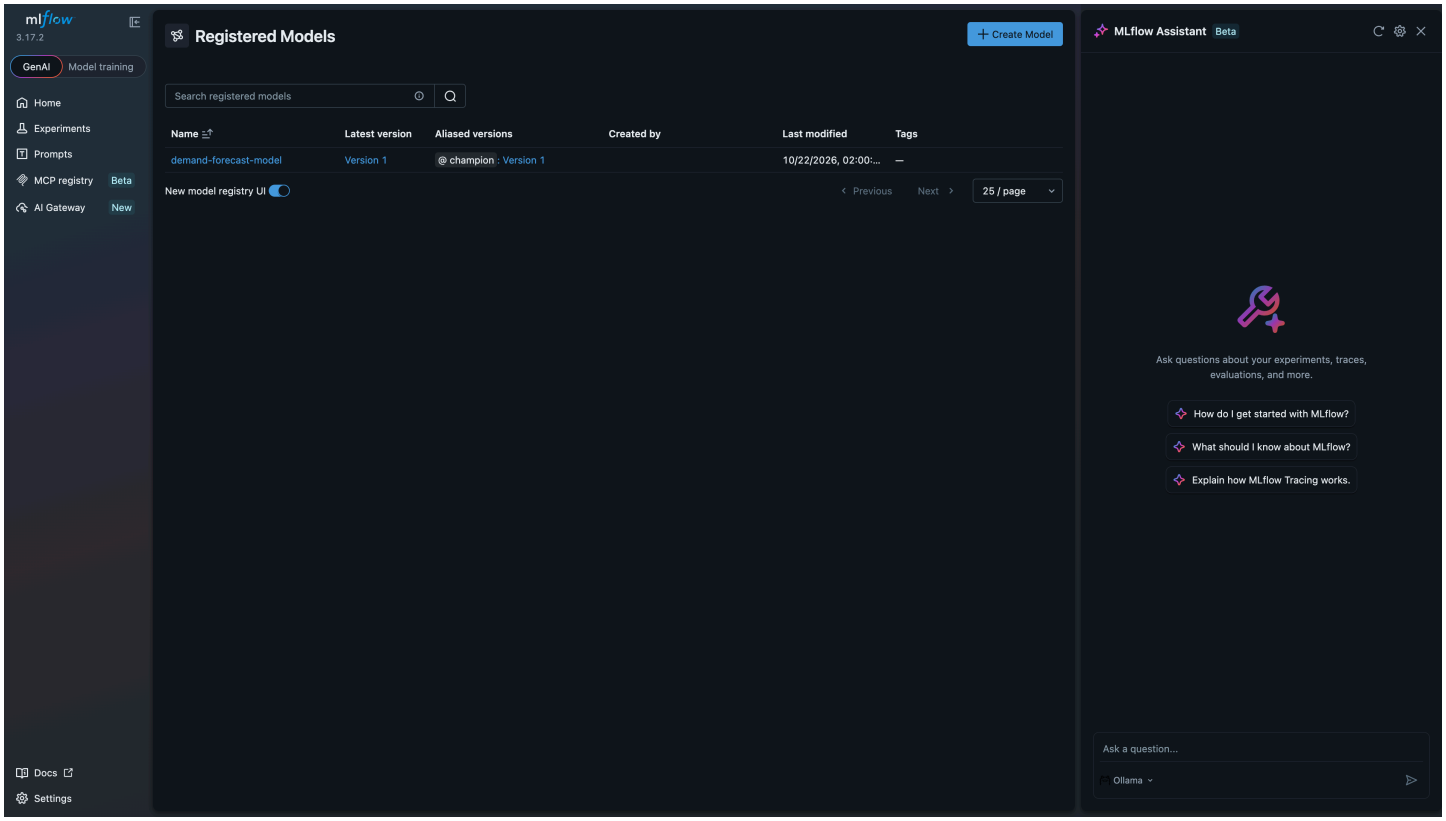
1. Prefect Flow Success - drift-triggered retrain completed



2. Evidently AI Drift Report - Wasserstein / dataset drift detected



3. MLflow Model Registry - demand-forecast-model v1 @champion



4. Email Alert Notification - retrain triggered

Email Alert Notification

```
From: drift-monitor@pipeline.local
To: ml-ops@example.com
Date: 2026-10-22T08:30:02.037909+00:00
Subject: [ALERT] Data drift detected — retrain triggered

Automated demand-forecasting pipeline alert.

Drift threshold (Wasserstein): 0.1
Max Wasserstein distance: 8.7455
Evidently report: outputs/drift_reports/drift_report.html

Per-column Wasserstein scores:
  - price: 6.0906
  - promo: 0.3045
  - day_of_week: 0.0485
  - temperature: 7.8507
  - store_id: 0.0550
  - units_sold: 8.7455

Retrain completed and model registered in MLflow:
  model_name: demand-forecast-model
  version: 1
  run_id: 2e20ec0d11774c8cbfa9689c0e29da77
  metrics: {'mae': 3.520356429576031 2, 'r2': 0.9529164538624942}
```

5. GitHub Repository - project uploaded

